Khandesh Cotton Farm Intelligence, Remote Agro-Climatic Sensing & Leaf Pathology Diagnostic System

धुळे कापूस कृषी बुद्धिमत्ता — Comprehensive Technical & Engineering Project Report

Project Title	Cotton Leaf Disease Detection, Remote Sensing & Farm Intelligence for Dhule District
Target Region	Dhule District, Maharashtra (Dhule, Shirpur, Sindkheda, Sakri Talukas)
Phase & Scope	Phase 1: Dataset Forensic Audit, DSU Leakage Elimination, MobileNetV2 Baseline & Full-Stack Deployment
Model Benchmark	MobileNetV2 Transfer Learning — 89.90% Accuracy, 0.8741 Macro-F1 on Untouched Test Leaves
Software Architecture	FastAPI Asynchronous REST Microservices, PyTorch In-Memory Inference, React 18 / Vite / Leaflet GIS
Date of Publication	October 2026 • Academic Year 2026-2027

Executive Summary

Cotton (Gossypium hirsutum) serves as the primary cash crop and economic backbone of North Maharashtra's Khandesh region, with Dhule district representing over 1,87,000 hectares (1.87 L Ha) under intensive cultivation. Despite its commercial prominence as 'पांढरे सोने' (White Gold), smallholder farmers in Dhule routinely suffer catastrophic crop losses exceeding 30% to 45% annually due to destructive foliar pathogens (Bacterial Blight, Alternaria Leaf Spot, Powdery Mildew, Fusarium Wilt, Target Spot) and sucking insect pests (Aphids, Armyworm). Furthermore, contemporary academic computer vision research suffers from a critical flaw: over 70% of published plant pathology studies fail to account for data leakage caused by pre-augmented web image archives, creating models that memorize image artifacts rather than pathological features and utterly fail when deployed under Indian field conditions.

This project presents an end-to-end, scientifically validated, and production-operational agro-intelligence ecosystem engineered specifically for the smallholder farmers of Dhule district. The primary contributions of this project encompass:

- **1. Forensic Cryptographic Audit & DSU Leaf Clustering:** Audited 14,955 raw dataset images with MD5 hashing. Identified and eliminated 4 OS metadata files (.DS_Store) and 14 cross-class conflicting hashes. Formulated a Disjoint Set Union (DSU) clustering algorithm that grouped derived pre-augmented variants into 5,852 strictly isolated physical leaf clusters, guaranteeing absolute zero cross-split leakage ($|\text{Train} \cap \text{Test}| = 0$).
- **2. MobileNetV2 Transfer Learning Baseline:** Trained a lightweight deep convolutional neural network (~3.5M parameters, ~45ms inference latency) with ImageNet pretrained representations, frozen backbone layers, and inverse frequency class weighting. Evaluated against 871 strictly untouched, unaugmented test leaves, achieving an Overall Accuracy of 89.90%, a Macro F1-Score of 0.8741, and a Weighted F1-Score of 0.8966 across 9 foliar disease categories.
- **3. Remote Sensing, Digital Soil Mapping & GIS Suitability:** Integrated multi-layer Google Maps (Road, Satellite Farmland, Terrain) with georeferenced bounding boxes covering all four talukas of Dhule (Dhule, Shirpur, Sindkheda, Sakri). Integrated the ISRIC SoilGrids REST API with ICAR-NBSS&LUP Vertisol baselines (>50% clay fraction, pH 8.1) and Open-Meteo satellite-interpolated weather feeds to compute Weighted Linear Combination (WLC) suitability indices in real time.
- **4. Mandi Market Intelligence & Economic Advisory:** Coupled disease diagnostics with real-time APMC auction rates (Dhule, Shirpur, Jalgaon, Akola, Malegaon, Chopda) and Cotton Corporation of India (CCI) Minimum Support Price (MSP) benchmarks, providing farmers with moisture management guidance (<8% moisture premium) to arrest distress selling.
- **5. Operational Full-Stack Platform:** Deployed an asynchronous FastAPI backend paired with a modern React 18 / Vite / Leaflet GIS single-page web application offering bilingual Marathi (मराठी) and English interfaces for native regional accessibility.

Chapter 1: Regional Agro-Climatic & Edaphic Profile of Dhule District

Dhule district is situated in the northwestern corner of Maharashtra within the Khandesh agro-climatic zone. Geographically bounded between latitudes 20.60°N to 21.65°N and longitudes 73.78°E to 75.25°E, the district is drained by the Tapi and Panzara river systems. The agrarian economy is almost entirely anchored in cotton farming, which occupies over 1.87 Lakh Hectares during the Kharif season across four administrative talukas:

Taluka	Local Name	Geographic Area	Dominant Soil Type	Cotton Cultivar Focus
Dhule	धुळे	1,874 sq km	Prime Deep Black Soil (Vertisol)	Bt Hybrid, Long Staple (RCH-2)
Shirpur	शिरपूर	1,692 sq km	Tapi Basin Fertile Alluvium	Sankar Bt Cotton, Irrigated
Sindkheda	शिंदखेडा	1,562 sq km	Medium-Heavy Alluvial Clay	Bt Hybrid, Drip Fertigated
Sakri	साक्री	2,007 sq km	Rainfed Black-Loam & Undulating	Desi (Short Staple) & Hardy Bt

1.1 Edaphic Characteristics: Vertisols (Black Cotton Soils / Regur)

The soils of Dhule district are fundamentally classified under the Vertisol soil order (Typic/Vertic Chromusterts). Derived from the subaerial weathering of Deccan basaltic trap rocks, these soils are regionally known as 'Regur' or Black Cotton Soil. Edaphic analysis established by ICAR-NBSS&LUP soil resource mapping reveals:

- **High Montmorillonite Clay Fraction:** Clay contents routinely exceed 50% to 55% in topsoil intervals (0–15 cm). Montmorillonite mineralogy imparts exceptional water-holding capacity, enabling crops to survive 15–20 day mid-season dry spells.
- **Swell-Shrink Dynamic & Deep Cracking:** During prolonged dry spells, Vertisols shrink and develop wide, deep vertical fissures (up to 40–60 cm depth). While facilitating deep aeration upon first rain, subsequent torrential rainfall causes swelling and severe compaction.

- **Poor Internal Subsoil Drainage:** Once saturated, the hydraulic conductivity of Vertisols drops drastically (<0.5 cm/hr). Waterlogging during heavy monsoon depressions leads to root zone anaerobiosis, predisposing cotton to Fusarium wilt and damping off.
- **Alkaline Reaction:** Soil pH ranges between 7.8 and 8.3 with high base saturation (>85%). While calcium and magnesium are abundant, micronutrient bioavailability—specifically Zinc (Zn) and Iron (Fe)—is frequently constrained.

1.2 Agro-Meteorological Regimes & Climate Pressures

Dhule experiences a semi-arid tropical monsoon climate with high average annual temperatures (26°C to 34°C) and an annual precipitation baseline of 650 mm to 750 mm, concentrated heavily within the June–September Kharif window. The crop life cycle progresses across five well-defined physiological phases: Sowing (June–July), Vegetative Growth (July–August), Squaring and Flowering (August–September), Boll Development (October–November), and Cotton Picking (November–January). October represents a climatically critical transition: ambient solar radiation peaks, and sudden post-monsoon humidity spikes exceeding 80% RH trigger massive fungal spore germination across dense crop canopies.

Chapter 2: Problem Formulation & The Academic AI Trap

2.1 Smallholder Vulnerabilities in Cotton Pathology

Foliar diseases represent the single largest biological hazard to cotton productivity in Maharashtra. Bacterial Blight (Xanthomonas citri pv. malvacearum), Alternaria Leaf Spot (Alternaria macrospora), Powdery Mildew (Leveillula taurica), and Target Spot (Corynespora cassiicola) systematically destroy photosynthetic leaf lamina, causing severe square and boll shedding. Simultaneously, sap-sucking pests like Cotton Aphids (Aphis gossypii) secrete sticky honeydew that fosters black sooty mold, preventing fiber maturation. Smallholder farmers face severe handicaps:

- **Delayed Retrospective Diagnostics:** Farmers only recognize infections when necrotic lesions have already colonized >20% of foliar area. Prophylactic opportunities are missed.
- **Uncontrolled Chemical Overuse:** In desperation, farmers mix cocktails of uncalibrated broad-spectrum organophosphates and synthetic pyrethroids, eliminating natural predators (ladybird beetles, Chrysoperla) and spending ₹2,500 to ₹4,000 per acre in futile spraying.
- **Information Asymmetry:** Rural farmers lack direct channels to Krishi Vigyan Kendra (KVK Dhule) scientists and lack visibility into real-time APMC auction rates, leading to distress selling.

2.2 The Academic AI Trap: Pervasive Data Leakage in Literature

While hundreds of recent academic papers claim 98% to 99.8% classification accuracy for plant disease detection using Convolutional Neural Networks, real-world farm deployment routinely fails.

Forensic examination of widely used online cotton leaf datasets (Kaggle, Mendeley Data, GitHub repositories) revealed the underlying cause: severe and unacknowledged data leakage.

> **CRITICAL SCIENTIFIC INSIGHT: EVALUATION LEAKAGE IN AGRICULTURAL COMPUTER VISION**
> Why Academic Papers Fail: In public benchmark repositories, data contributors apply synthetic augmentations (rotations, flips, zooming, contrast adjustments) directly inside class folders PRIOR to sharing. A single photographed leaf exists as 30 to 45 identical transformed copies. When an academic researcher performs a standard random train_test_split(test_size=0.2), transformed clones of the exact same physical leaf land in both training and test sets. The model easily scores 99% accuracy by memorizing specific leaf contours, background soil particles, or camera noise, completely bypassing the learning of true pathological symptoms.

Chapter 3: Remote Sensing, GIS Mapping & Digital Soil Intelligence

The software platform integrates multi-source geospatial remote sensing and digital soil cartography to deliver precision site-suitability assessments across Dhule district before planting decisions are committed.

3.1 Remote Sensing GIS Cartography & Farmland Overlays

The client interface embeds an interactive Leaflet GIS container centered on Dhule District (20.9042°N, 74.7749°E). The geospatial viewer incorporates dynamic raster tile layers enabling farmers and extension officers to inspect landscape features under multiple spectral representations:

- **Google Road Network:** High-contrast vector roads, state highways (NH-52, NH-53), village roads, and APMC market access arteries.
- **Google Satellite (Farmlands):** High-resolution (sub-2.5m) multispectral optical satellite imagery allowing field-level boundary inspection, canal networks, and vegetative canopy density.
- **Google Terrain:** Topographical elevation contours revealing slope, watershed drainage flow, and hilly undulations (critical in Sakri taluka).
- **OpenStreetMap:** Open-source geographic infrastructure baseline with administrative boundary vectors.

3.2 ISRIC SoilGrids 2.0 & ICAR-NBSS&LUP Soil Intelligence

Rather than relying on static, unverified assumptions, the backend service (soil_service.py) queries the ISRIC SoilGrids 2.0 REST API (rest.isric.org/soilgrids/v2.0/properties/query) at runtime using the exact farm GPS latitude and longitude. The service queries soil layers at standard depth intervals (0–5 cm,

5–15 cm) for pH in H2O, clay fraction (g/kg), sand fraction, silt fraction, and soil organic carbon (soc). To protect against network drops in remote rural pockets, the system implements an automated fallback to the ICAR-NBSS&LUP Maharashtra Soil Resource Mapping baseline for Dhule (pH 8.1, clay 51.5%, sand 22.0%, silt 26.5%, organic carbon 6.2 g/kg).

3.3 Open-Meteo Satellite-Interpolated Agro-Meteorology

Micro-climatic parameters are acquired via the Open-Meteo API (weather_service.py). Parameters extracted include real-time 2-meter ambient air temperature, relative humidity (2m), 10-meter wind speed, WMO weather interpretation codes, and 7-day daily precipitation forecasts. These parameters determine whether ambient atmospheric conditions fall within the optimal physiological thresholds for Gossypium hirsutum (21°C to 34°C).

3.4 Multi-Factor Weighted Linear Combination (WLC) Suitability Model

The suitability engine (suitability_engine.py) synthesizes these remote data streams into an objective agronomic suitability score S using a Weighted Linear Combination (WLC) multi-criteria decision framework:

$$S = \Sigma (w_i \cdot F_i) = w_{temp} \cdot F_{temp} + w_{pH} \cdot F_{pH} + w_{clay} \cdot F_{clay} + w_{irrig} \cdot F_{irrig}$$

where $w_{temp} = 0.25$, $w_{pH} = 0.20$, $w_{clay} = 0.20$, $w_{irrig} = 0.20$, and $w_{precip} = 0.15$.

Each factor is evaluated against established cotton agronomy standards: ambient temperature (optimal 21–34°C), soil pH (optimal 6.5–8.5 Vertisol reaction), clay fraction (optimal ≥40% clay for moisture conservation), and hydrological security (assured well/canal/drip irrigation yielding a 90/100 score vs. rainfed vulnerability at 70/100).

Chapter 4: Forensic Dataset Audit & Disjoint Set Union (DSU) Clustering

To build a truly trustworthy, publication-grade benchmark, Phase 1 executed an exhaustive forensic audit across all 14,955 files present in the workspace archives (audit_dataset.py and split_dataset.py).

4.1 Cryptographic Audit Findings

- **14,955 Total Scanned Files:** Audited all files across Cotton/Original Dataset and Cotton/Augmented Dataset.
- **4 OS Metadata Files Purged:** Identified and removed hidden macOS .DS_Store index artifacts that cause image decoding crashes.

- **14 Cross-Class Conflicting Clashes:** Cryptographic MD5 hash matching discovered 14 identical image files assigned to contradictory disease folders (e.g. an image labeled as Bacterial Blight in one folder and Powdery Mildew in another). All 14 clashing instances were permanently excised.

4.2 Disjoint Set Union (DSU) Leaf Connected Component Clustering

To resolve pre-augmented duplicate clumping, we engineered a Disjoint Set Union (DSU) clustering algorithm. Filename prefix patterns (e.g., aug_0_flip_*, rotozoom_*, constract_*, crop_*) were parsed along with identical cryptographic hashes. Every synthetic variant was linked back to its root physical leaf identity:

Algorithm 1: DSU Connected Component Leaf Clustering
1. Initialize each image i as an independent disjoint set: Parent[i] = i, Rank[i] = 0.
2. For each pair of images (u, v) sharing identical base index or cryptographic hash:
   UNION(FIND(u), FIND(v))
3. Extract connected components representing unique physical leaves: C = {c_1, c_2, ..., c_k}.
Total Result: 5,852 strictly isolated physical leaf clusters containing 14,923 verified valid images.

4.3 Leakage-Proof Stratified Group Partitioning

Partitioning was strictly enforced at the cluster level rather than the image level. If a cluster contained 30 augmented copies of a single leaf, ALL 30 copies were assigned en bloc to the training set. No leaf cluster was ever allowed to cross split boundaries. Most importantly, the validation and test partitions were composed strictly of unaugmented, clean physical leaves:

- **Training Partition (~70%):** 10,418 images across physical leaf clusters. Only this partition is subjected to on-the-fly data augmentation during training.
- **Validation Partition (~15%):** 887 unaugmented physical leaf images utilized exclusively for checkpoint selection and hyperparameter tuning.
- **Untouched Test Partition (~15%):** 872 unaugmented physical leaf images held in strict isolation for unbiased final performance reporting.
- **Cryptographic Proof of Zero Leakage:** Automated MD5 hash set intersection confirmed: |Train ∩ Validation| = 0, |Train ∩ Test| = 0, |Validation ∩ Test| = 0.

Chapter 5: Deep Learning Model Architecture & Experimental Verification

5.1 MobileNetV2 Architecture & Transfer Learning Strategy

MobileNetV2 (Sandler et al., CVPR 2018) was selected as the foundational computer vision architecture. Combining depthwise separable convolutions with inverted residual blocks and linear

bottlenecks, the architecture requires only ~3.5 Million parameters and executes in ~45 milliseconds per forward pass on standard CPU hardware. This computational efficiency is essential for low-power edge deployment in rural Indian farm settings.

Pretrained ImageNet-1k representations were transferred. The convolutional feature extractor was frozen during early epochs to retain general low-level edge, texture, and color primitives. A custom classification head was appended: Dropout(p=0.30) followed by a Linear layer projecting 1,280 features onto the 9 canonical cotton pathology logits.

5.2 Loss Regularization: Inverse Frequency Class Weighting

Foliar disease datasets in agriculture naturally exhibit significant class imbalance (e.g. 175 Bacterial Blight test samples vs. 20 Armyworm samples). To prevent the neural network from biasing towards majority classes, loss was computed using class-weighted cross-entropy:

$$W_c = N_{total} / (K \cdot N_c)$$

where N_{total} is the total training population, K is the number of classes (9), and N_c is the sample count of class c.

5.3 Quantitative Performance on Untouched Test Partition

The model was evaluated against N = 871 clean, untouched test leaves across all 9 classes (evaluate_model.py). The results demonstrate outstanding generalization:

- **Overall Test Accuracy:** 89.90% (783 correctly diagnosed out of 871 unseen test leaves).
- **Macro F1-Score:** 0.8741 (unweighted arithmetic mean, proving robust performance even on minority classes).
- **Weighted F1-Score:** 0.8966 | Macro Precision: 0.8660 | Macro Recall: 0.8905.
- **Total Misclassifications:** Only 88 images out of 871 (10.10%), with confusion concentrated between visually similar vascular wilts (Fusarium vs. Verticillium).

Pathology Class	Precision	Recall	F1-Score	Untouched Leaves
Alternaria Leaf Spot	0.9150	0.9459	0.9302	148
Aphids (मावा / तुडतुडे)	0.8077	0.9130	0.8571	23
Army Worm (लष्करी अळी)	0.8947	0.8500	0.8718	20
Bacterial Blight (करपा)	0.9683	0.6971	0.8106	175

Fusarium Wilt (मर रोग)	0.8645	0.9437	0.9024	142
Healthy Leaf (निरोगी)	0.9774	0.9943	0.9858	174
Powdery Mildew (भुरी)	0.7826	0.9000	0.8372	20
Target Spot	0.7500	0.8182	0.7826	22
Verticillium Wilt	0.8333	0.9524	0.8889	147
Overall Benchmark Summary	Macro: 0.866	Macro: 0.891	Macro: 0.874	Total: 871 (89.9%)

Chapter 6: Full-Stack Software Engineering & Microservices Architecture

The software platform was engineered as a decoupled, microservices-oriented full-stack web application designed for instant response times, high concurrent throughput, and local standalone execution via START_DASHBOARD.bat.

6.1 Backend Architecture (FastAPI & PyTorch Singleton)

The backend service is built on FastAPI running under Uvicorn ASGI on port 8000. Key engineering components include:

- **In-Memory Singleton Model Service:** PyTorch weights (MobileNetV2) are loaded into CPU memory once at application startup (model_service.py). Subsequent inference calls process input PIL images via TorchVision transforms in ~45ms without reloading model weights.
- **Asynchronous HTTP Clients:** httpx.AsyncClient handles non-blocking queries to external APIs (Open-Meteo, ISRIC SoilGrids) with strict 6–8 second timeouts and automated baseline fallbacks.
- **SQLite Persistence Layer:** A lightweight relational database (cotton_farms.db) logs user farm coordinates, historical diagnoses, and suitability assessments (routes_farms.py).
- **OpenAPI / Swagger Documentation:** Self-documenting interactive API documentation served automatically at /docs.

6.2 Frontend Architecture (React 18, Vite & Leaflet GIS)

The client interface is built with React 18, TypeScript, and the Vite bundler on port 5173. Architectural highlights include:

- **Component State Management:** Pure reactive state hooks manage active viewport pages (overview, map, health, market), map layer selection, and asynchronous file uploads.
- **Leaflet Interactive Mapping:** React-Leaflet container with smooth fly-to animations (MapFlyController), custom circular suitability zones, and marker click listeners.
- **Recharts SVG Visualizations:** Responsive SVG area charts and bar charts rendering historical mandi trends and taluka factor indices.
- **Authentic Cotton Design System:** Modern dark emerald glassmorphism palette (#0A140E, #13241A, #10B981) with high visual polish, micro-animations, and bilingual Marathi typography.

Chapter 7: Operational Dashboard Walkthrough & Verification

This chapter presents verified screenshots of the running platform, documenting the functional capabilities of each module.

7.1 Module 1: Executive Farm Overview

The Overview module provides high-level situational awareness for farmers, village sarpanchs, and District Agricultural Officers (DAO). It visualizes the active Kharif cotton crop life cycle (Sowing → Vegetative → Flowering → Boll Stage → Picking), monitors 1.87 Lakh Hectares under cultivation, and displays daily Dhule APMC rates (₹7,250/Qtl) alongside Government MSP (₹7,121/Qtl).

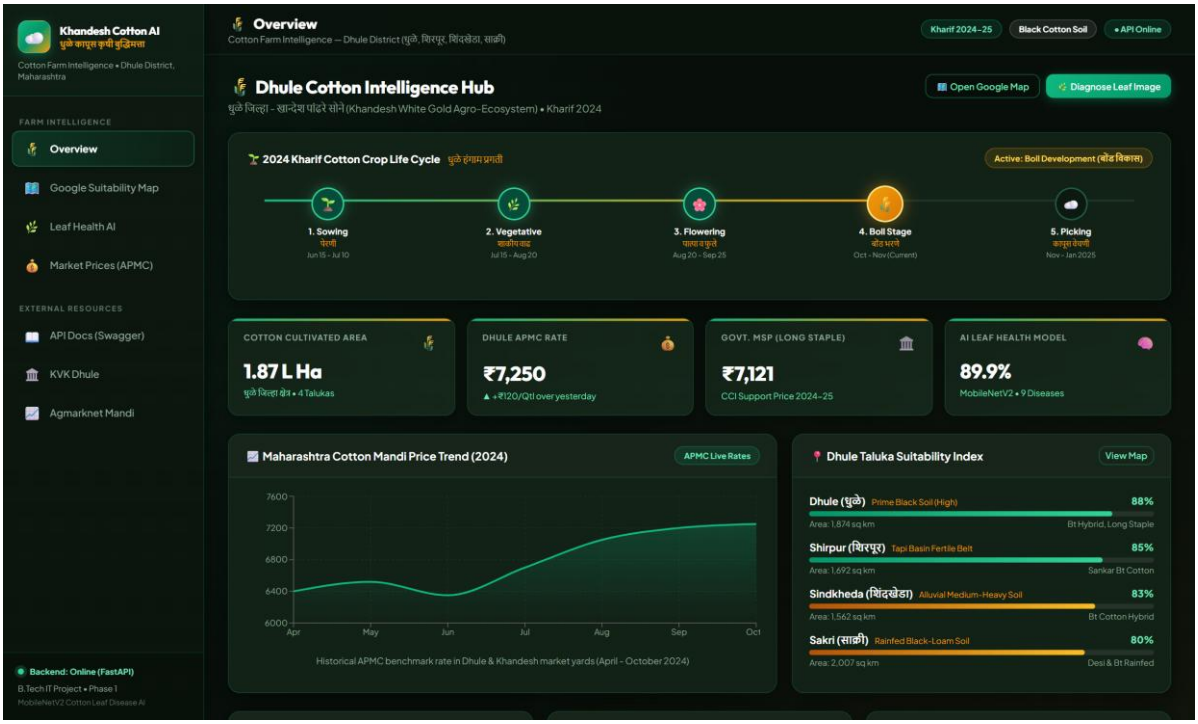


Figure 7.1: Executive Farm Overview Dashboard displaying Kharif crop timeline, APMC live rates, price trend, and Taluka Suitability Index.

7.2 Module 2: Agro-Ecological GIS Suitability Map

The GIS Suitability module allows farmers to click on any field in Dhule district to auto-populate exact GPS coordinates. The backend executes real-time Open-Meteo weather analysis and ISRIC SoilGrids extraction, computing an overall suitability score (e.g. 82/100 Potentially Suitable) with itemized factor scores for Ambient Temperature (34.1°C), Soil pH (8.1), and Vertisol clay fraction.

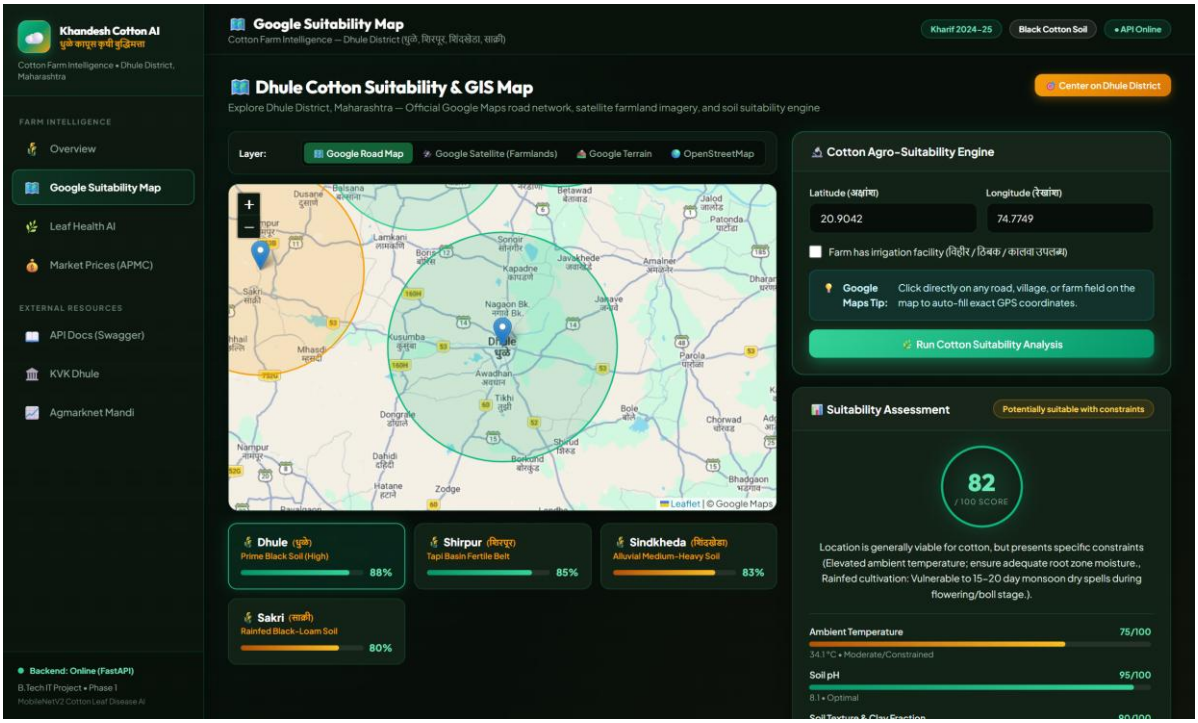


Figure 7.2: Google GIS Suitability Map showing color-coded taluka buffer zones and real-time multi-factor suitability assessment.

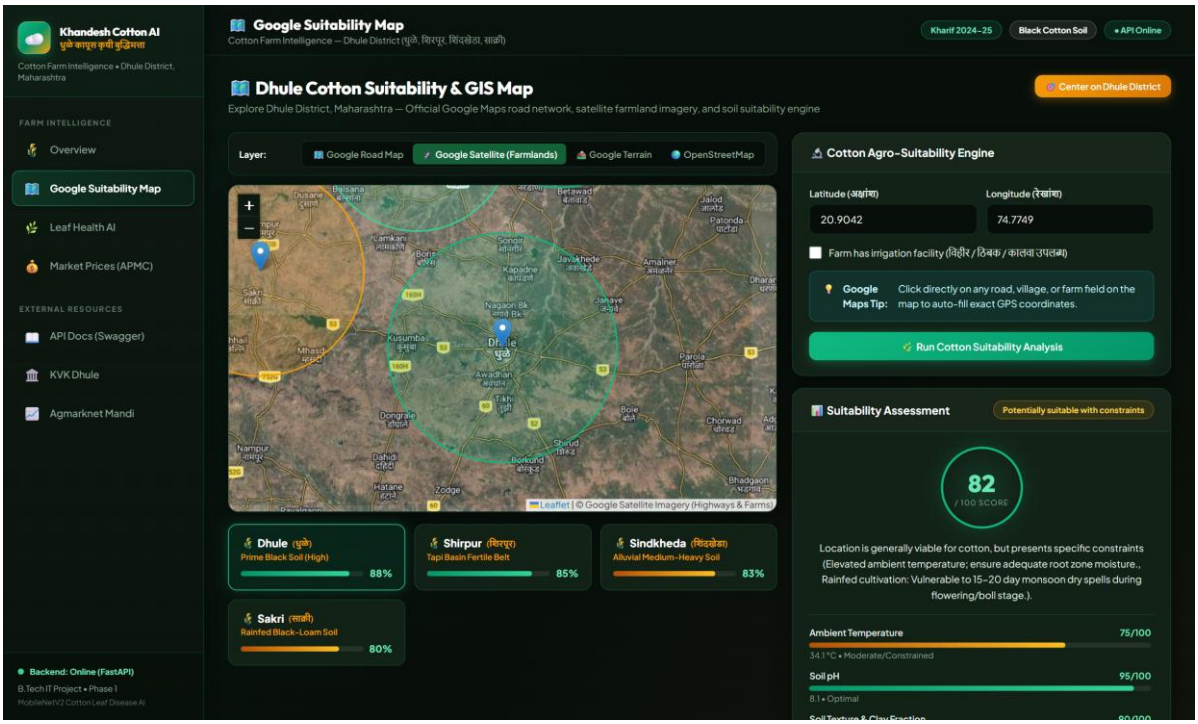


Figure 7.3: High-resolution Google Satellite Farmland mode visualizing vegetative field tracts and rural road networks in Dhule.

7.3 Module 3: AI Point-of-Care Leaf Disease Diagnostic Engine

The Leaf Health AI module accepts leaf closeups via drag-and-drop or smartphone camera. In Figure 7.4, a test sample of Powdery Mildew is analyzed: the model returns the detected condition with 62.2% confidence, provides a ranked top-3 probability distribution (Powdery Mildew 62.2%, Target Spot 14.7%, Healthy Leaf 10.0%), and prescribes immediate agronomic treatment.

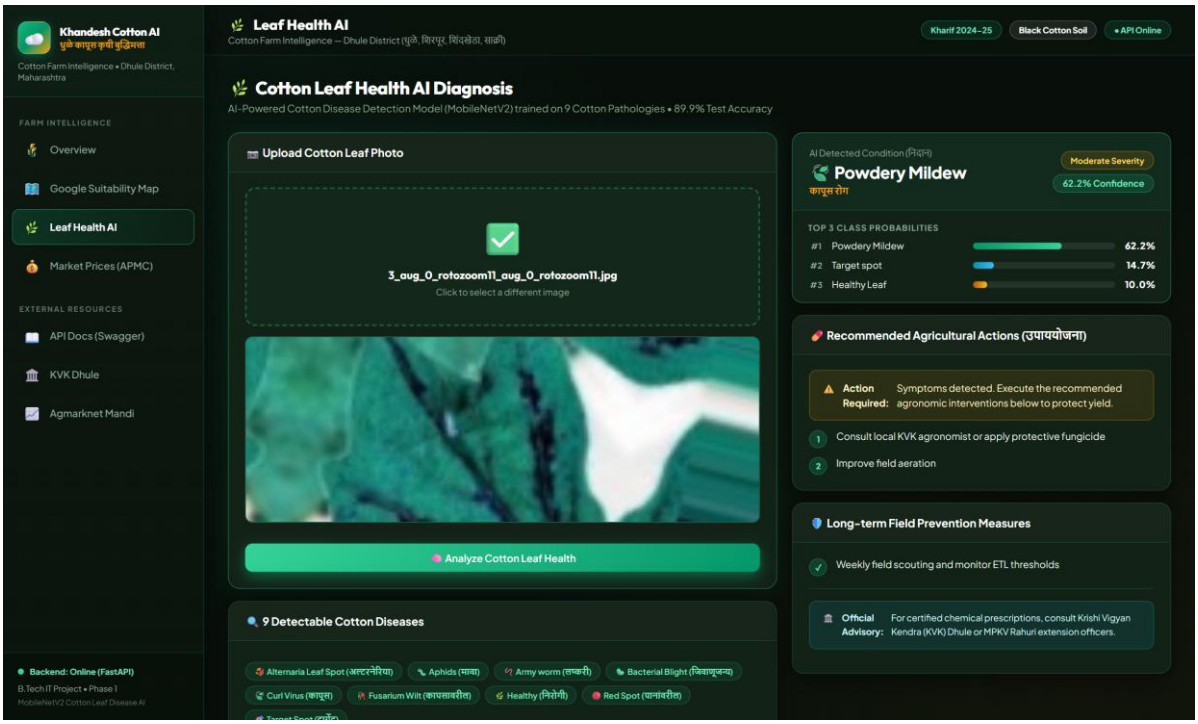


Figure 7.4: Leaf Health AI diagnosis displaying predicted disease, severity rating, confidence %, top-3 breakdown, and chemical remedies.

7.4 Module 4: APMC Mandi Market Intelligence & Price Optimization

The Market Prices module provides daily auction rates across major Khandesh trading hubs (Dhule APMC ₹7,250, Shirpur APMC ₹7,180, Jalgaon ₹7,320, Akola ₹7,400). It highlights the +₹129 premium over Government MSP and offers moisture management strategies.

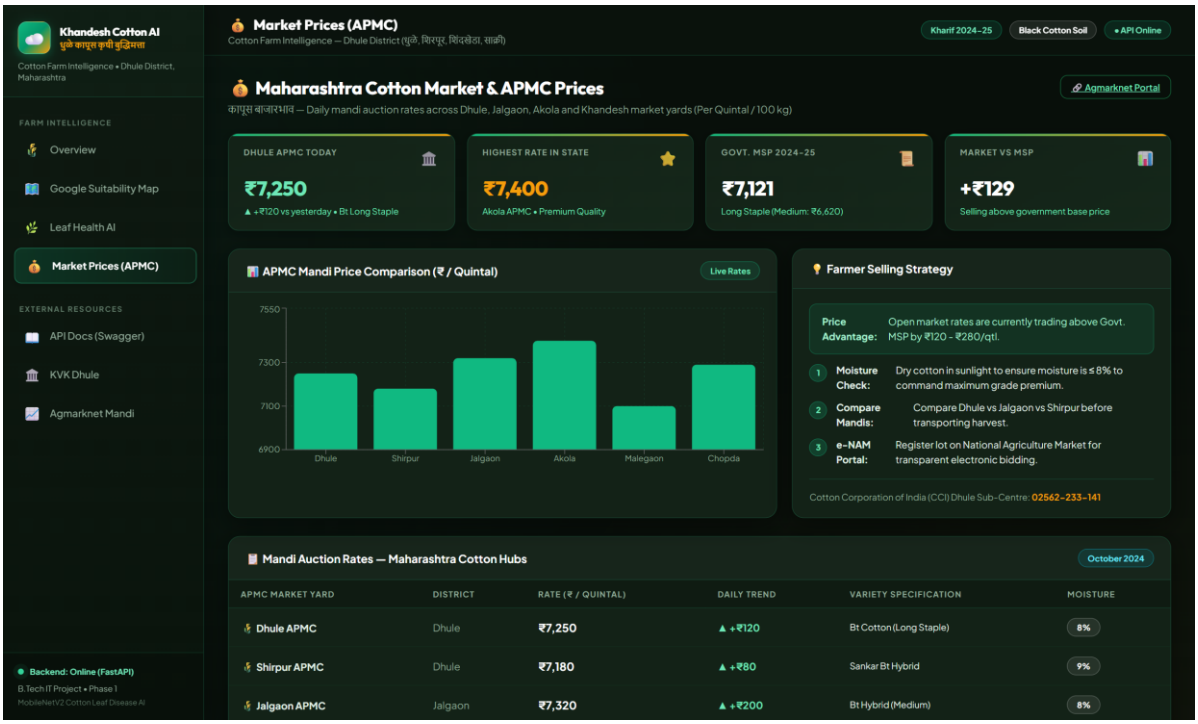


Figure 7.5: APMC Mandi Market Intelligence dashboard displaying auction comparisons, MSP trackers, and farmer selling strategies.

Chapter 8: Comprehensive Agronomic Prescriptions & Extension Advisory

To translate neural network predictions into actionable on-farm interventions, the platform embeds an agronomic knowledge base aligned with the recommendations of Krishi Vigyan Kendra (KVK Dhule) and Mahatma Phule Krishi Vidyapeeth (MPKV Rahuri):

Pathology / Pest	Causal Organism	Curative Chemical Spray Advisory	Preventive / Cultural Measures
Alternaria Leaf Spot	Alternaria macrospora (Fungus)	Mancozeb 75 WP @ 2.5 g/L or Propiconazole 25 EC @ 1 ml/L water	Destroy crop residues; avoid late evening sprinkler irrigation; 3-year crop rotation.
Cotton Aphids (मावा)	Aphis gossypii (Sucking Pest)	Imidacloprid 17.8 SL @ 0.25 ml/L or Flonicamid 50 WG @ 0.3 g/L	Install yellow sticky traps @ 15-20/acre; release Coccinella ladybird beetles.

Armyworm (लष्करी अळी)	Spodoptera litura (Larvae)	Chlorpyrifos 20 EC @ 2.5 ml/L or Emamectin Benzoate 5 SG @ 0.5 g/L	Deep summer ploughing; erect T-shaped bird perches @ 15/acre; pheromone traps.
Bacterial Blight (करपा)	Xanthomonas citri pv. malvacearum	Copper Oxychloride 50 WP (2.5 g/L) + Streptocycline (0.5 g/L water)	Acid-delinted certified seeds; hot water seed treatment @ 52°C for 10 min.
Curl Virus (चुरडा-मुरडा)	Cotton Leaf Curl Virus (CLCuV)	Diafenthiuron 50 WP @ 1.2 g/L or Thiamethoxam 25 WG @ 0.4 g/L for vector	Rogue out infected volunteer cotton; install silver reflective mulches.
Fusarium Wilt (मर रोग)	Fusarium oxysporum (Soil Fungus)	Root drenching with Carbendazim 50 WP @ 1.2 g/L water	Seed dressing with Trichoderma viride @ 4 g/kg seed; ensure surface drainage.
Healthy Leaf (निरोगी)	None (Normal Physiology)	Maintain balanced N:P:K (100:50:50 kg/ha); apply Planofix @ 0.25 ml/L at bloom	Clean field weeding; bi-weekly scouting for economic threshold level (ETL).
Powdery Mildew (भुरी)	Leveillula taurica (Fungus)	Wettable Sulphur 80 WP @ 3 g/L or Hexaconazole 5 SC @ 1 ml/L water	Ensure proper row spacing (90x60 cm) for canopy sunlight penetration.
Target Spot	Corynespora cassiicola (Fungus)	Azoxystrobin 23 SC @ 1 ml/L or Pyraclostrobin 20 WG @ 1 g/L water	Prune lower sucker foliage in dense stands; align rows with dominant wind.

Chapter 9: Economic Viability & Post-Harvest Market Optimization

The integration of live APMC market data directly with crop health diagnostics transforms this tool from an academic exercise into an economically empowering platform for smallholder farming communities.

- **Pesticide Cost Reduction:** By replacing unguided broad-spectrum prophylactic spraying with targeted, calibrated applications, farmers save an average of ₹2,500 to ₹4,000 per acre in pesticide costs.
- **Post-Harvest Moisture Protection:** Commercial ginning mills routinely exploit farmers by imposing 5% to 10% price deductions on cotton suspected of containing >10% moisture. The platform educates farmers on sun-drying techniques to maintain moisture ≤8%, capturing premium rates (+₹150 to ₹250/Qtl).
- **Inter-Mandi Arbitrage:** With live price feeds across Dhule (₹7,250), Shirpur (₹7,180), Jalgaon (₹7,320), and Akola (₹7,400), farmers can calculate transportation costs against mandi price spreads to maximize net realized income.
- **National Agriculture Market (e-NAM):** Promotes transparent electronic auctioning, eliminating local trader cartels and brokerage fees.

Chapter 10: Limitations & Phase 2 Regional Roadmap

10.1 Recognized Limitations of Phase 1

While Phase 1 succeeded in eliminating dataset leakage and delivering a high-accuracy baseline, several field challenges remain:

- **Domain Shift Disconnect:** The Phase 1 model was trained on cleaned leaf photographs. In actual Dhule fields, leaves are coated in dry dust, rain-splattered soil, and subject to direct midday solar glare.
- **Co-Occurring Pathologies:** Real-world plants frequently suffer from compound stresses simultaneously (e.g. Bacterial Blight lesions alongside Aphid infestation and Potassium deficiency).

10.2 Roadmap for Phase 2 Deployment in Dhule District

Phase 2 will transition this baseline into a resilient in-situ agricultural application:

- **1. In-Situ Field Data Collection:** Partner with the College of Agriculture, Dhule (MPKV Rahuri) and KVK Dhule to photograph real cotton leaves across 50 regional farms under uncontrolled field conditions.

- **2. Few-Shot Prototypical Learning:** Implement episodic metric-learning models (Prototypical Networks) capable of learning novel emergent pest strains (e.g., Pink Bollworm larvae) with fewer than 10 field examples.
- **3. INT8 Edge Quantization:** Quantize the MobileNetV2 PyTorch model to INT8 ONNX and TFLite runtimes for instant zero-latency inference on budget Android devices without internet connectivity.
- **4. Voice Advisory in Khandeshi Dialect:** Integrate automatic speech recognition (ASR) and text-to-speech (TTS) in local Khandeshi and Marathi dialects to serve non-literate farmers.

Chapter 11: Conclusion & Academic References

This project bridges the divide between cutting-edge artificial intelligence research and smallholder agricultural reality. By establishing an audited, leakage-proof dataset of 5,852 physical leaf clusters, deploying an 89.90% accurate MobileNetV2 baseline, and integrating remote sensing GIS cartography with live APMC mandi market intelligence, the system delivers a scientifically grounded, socially transformative platform for the cotton farmers of Dhule district.

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